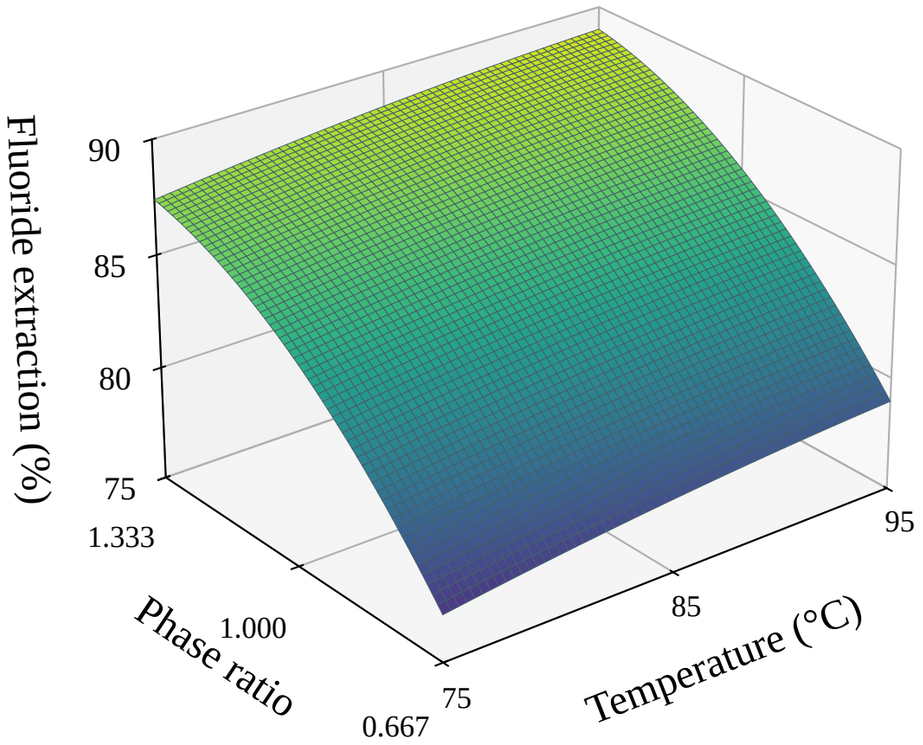
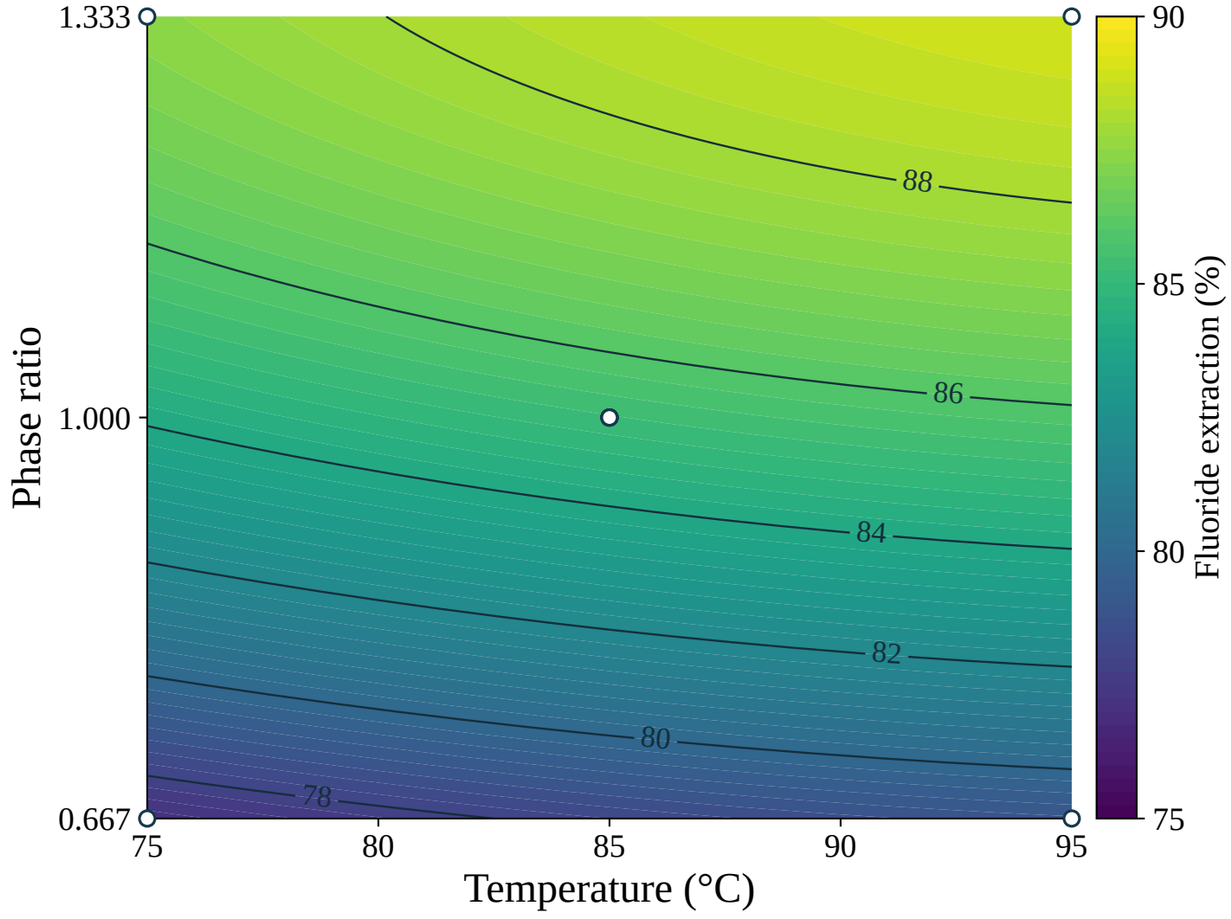
Fluoride extraction | Initial fluoride = 1.0%

(a) Response surface



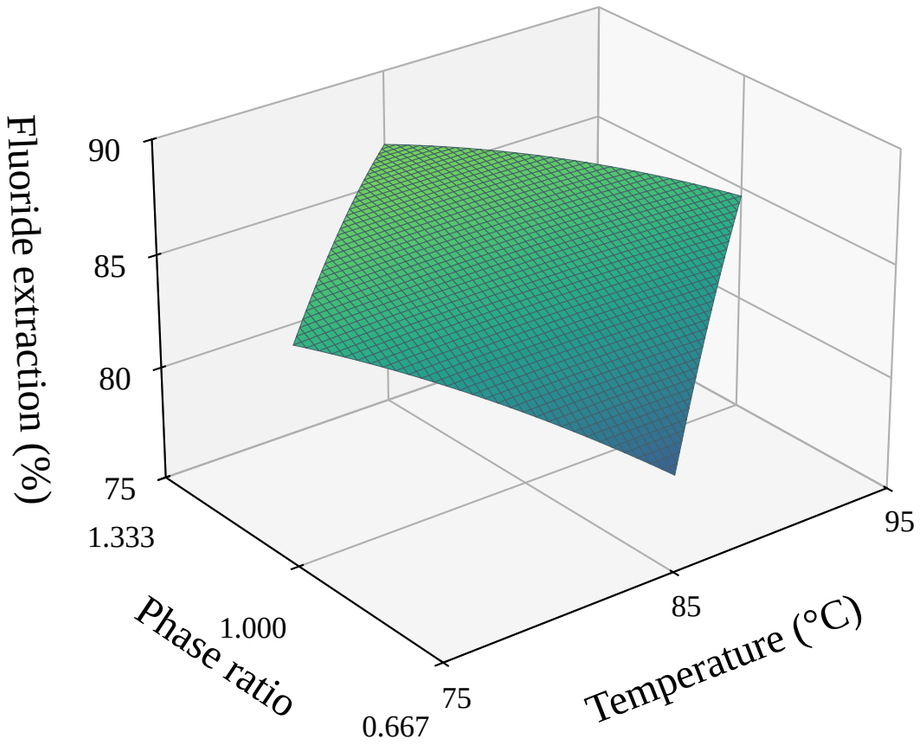
(b) Contours; circles = measured settings



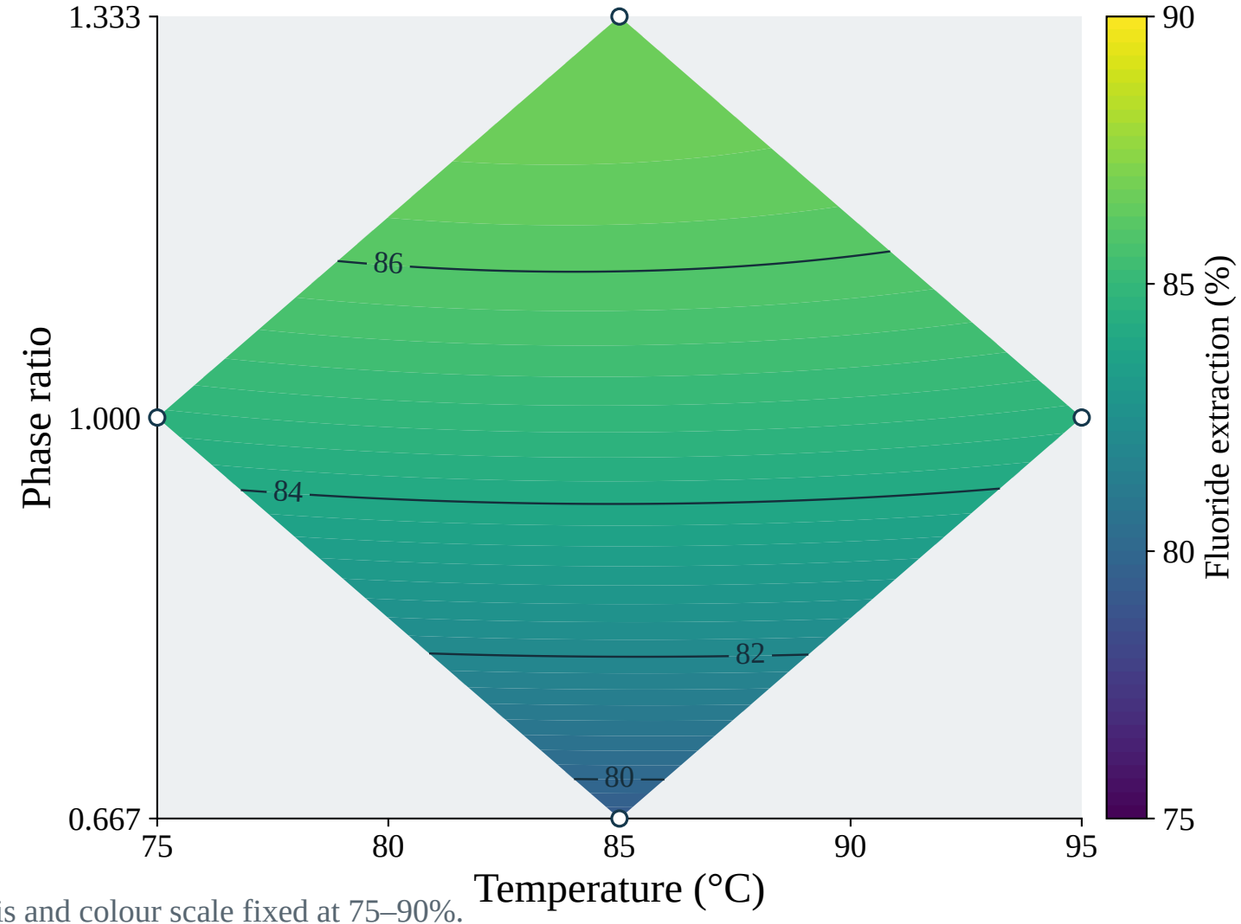
Saved quadratic model; vertical axis and colour scale fixed at 75–90%.

Fluoride extraction | Initial fluoride = 0.8%

(a) Response surface

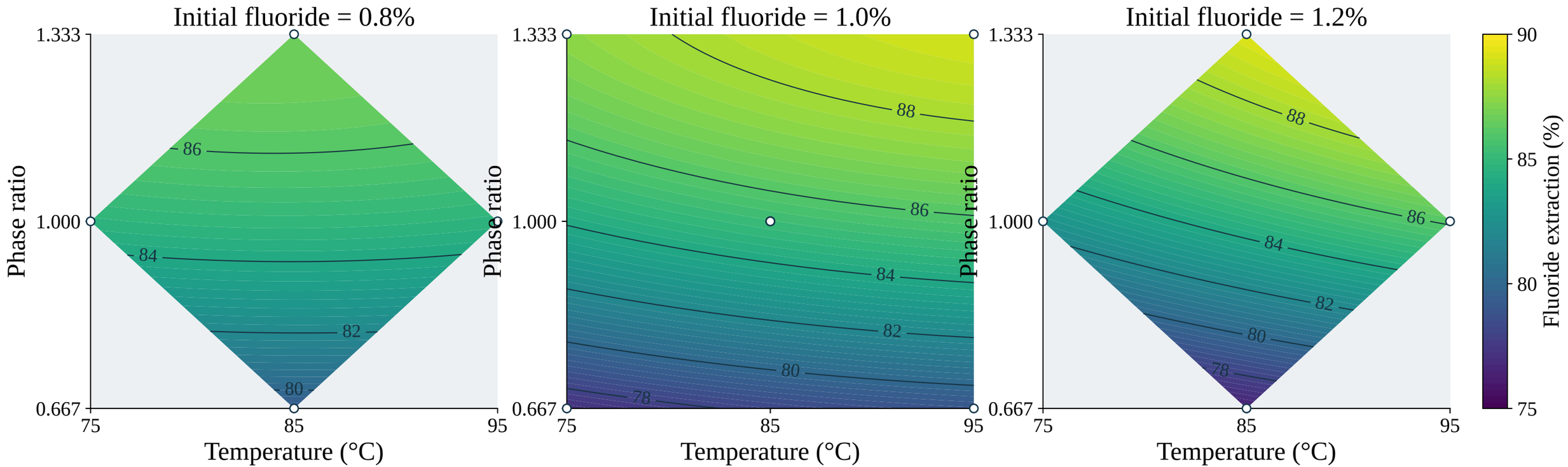


(b) Contours; circles = measured settings



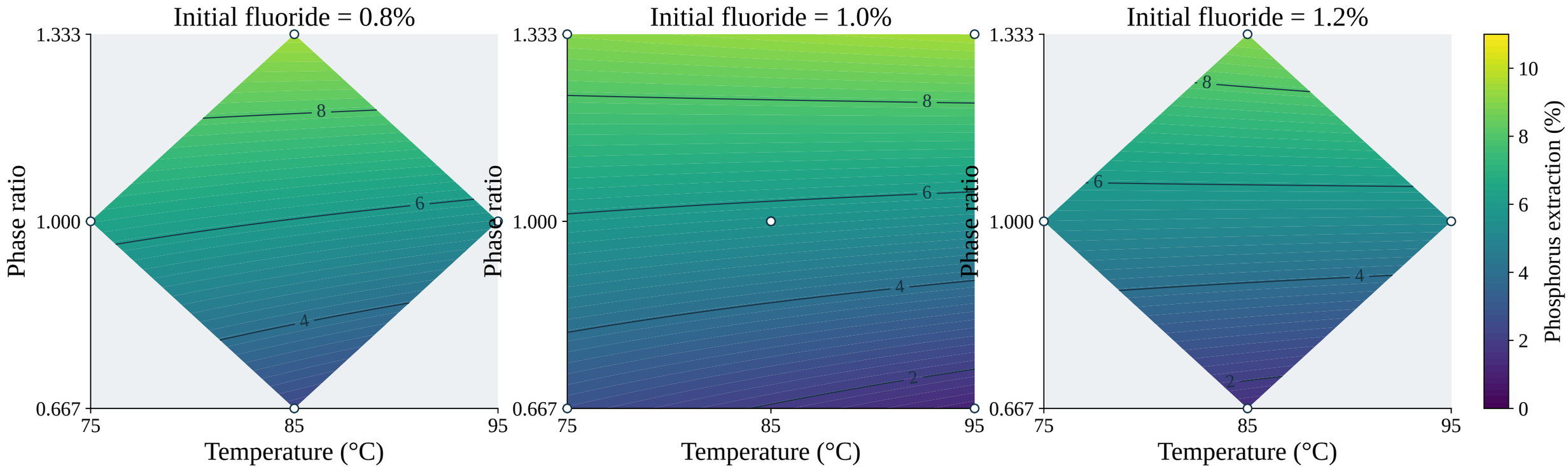
Saved quadratic model; vertical axis and colour scale fixed at 75–90%.
Grey corners are outside the convex hull of the experimental settings.

Fluoride extraction (%) | saved quadratic model



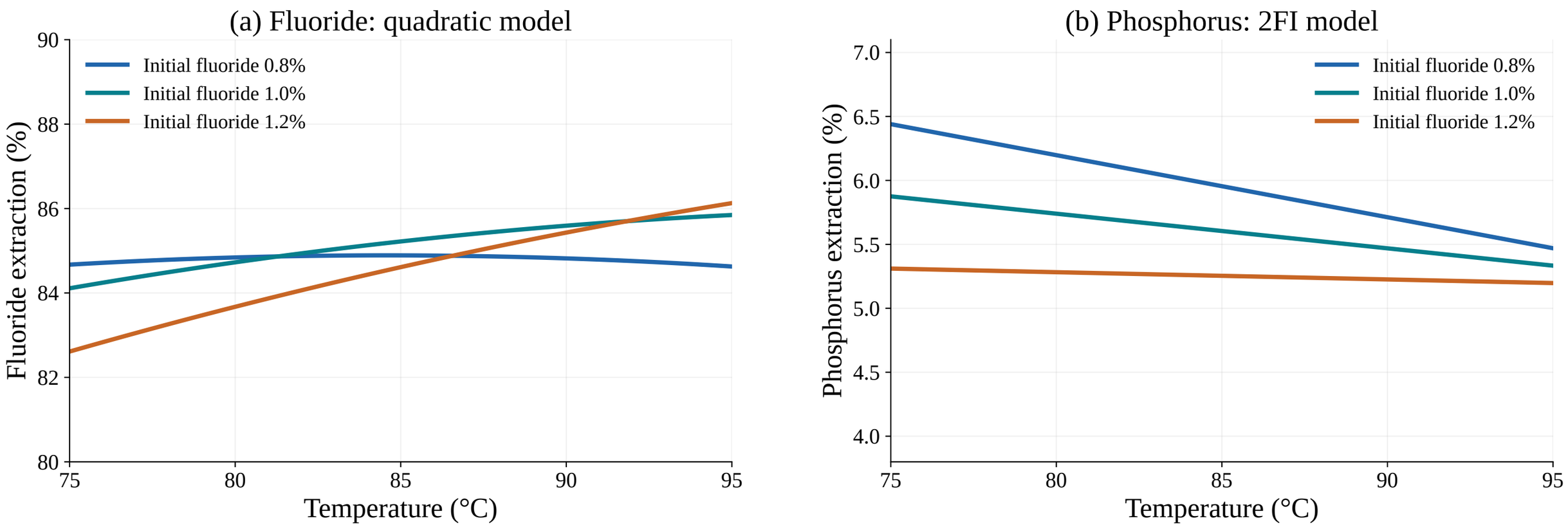
Identical axes and colour scales across panels. Circles = measured settings.
Grey corners = outside the experimental convex hull; coloured regions show model predictions.

Phosphorus extraction (%) | saved 2FI model



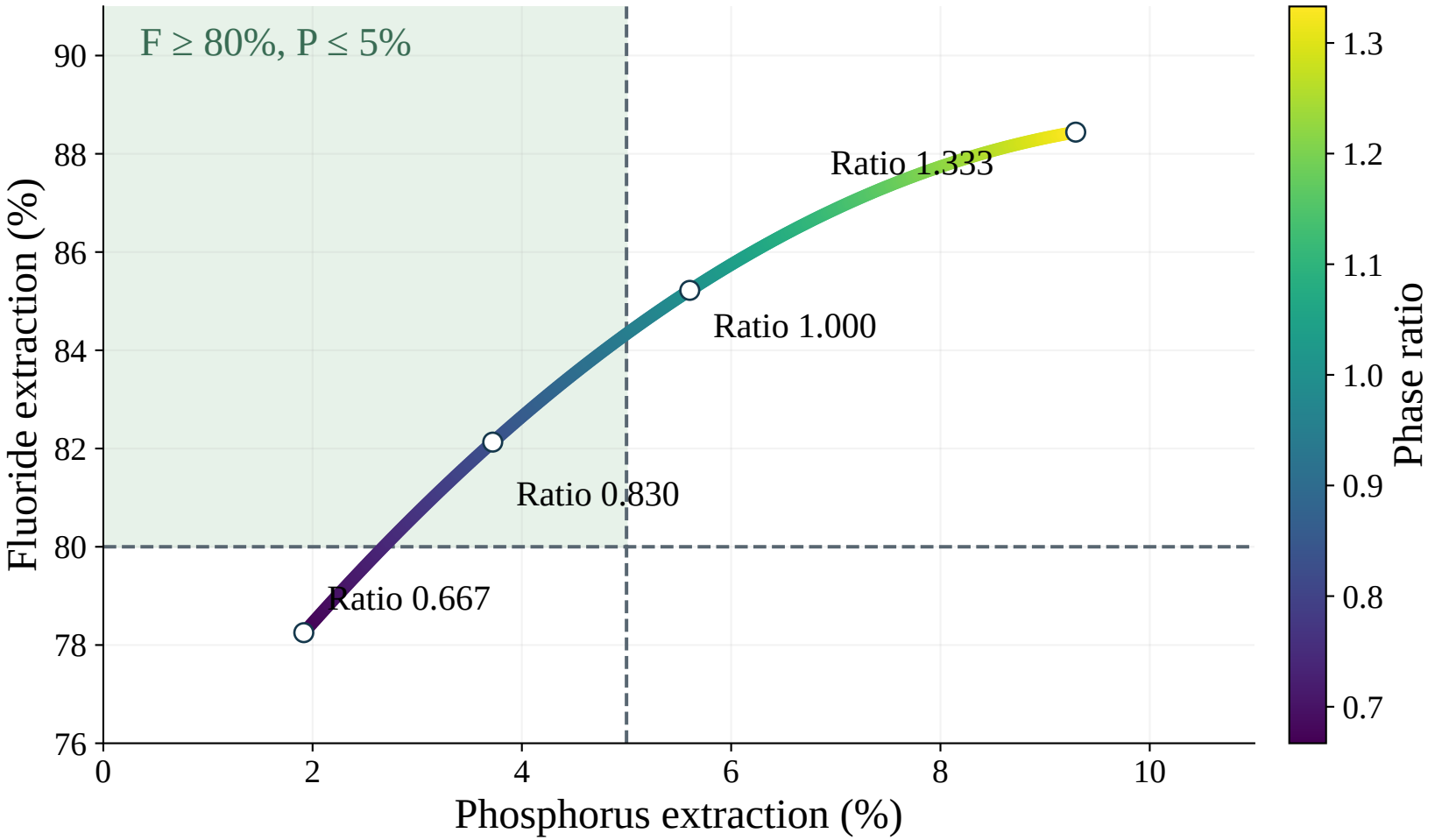
Identical axes and colour scales across panels. Circles = measured settings.
Grey corners = outside the experimental convex hull; coloured regions show model predictions.

Temperature response depends on initial fluoride | Phase ratio = 1.000



Conditional mean predictions from the unchanged project models.
The panels use different vertical scales; slopes alone do not establish statistical significance.

Increasing phase ratio: fluoride recovery and phosphorus loss



Temperature = 85°C; initial fluoride = 1.0%. Curve = model mean predictions.
The shaded target region does not imply that each future experimental batch will comply.